

ANN1: Activation Functions – Detailed Code & Viva Explanation

Cell 1: Import Libraries

Code:

```
import numpy as np
import matplotlib.pyplot as plt
```

Explanation (what to say in viva):

Here I am importing numpy and matplotlib libraries.

Numpy is used for numerical and array operations.

Matplotlib is used for plotting graphs.

Graphs help visualize activation functions.

Cell 2: Input Data

Code:

```
x = np.linspace(-10, 10, 100)
```

Explanation (what to say in viva):

This creates 100 evenly spaced values between -10 and 10.

These values act as input to activation functions.

It helps in plotting smooth curves.

Cell 3: Linear Function

Code:

```
def linear(x):
    return x
```

Explanation (what to say in viva):

This is a linear activation function.

It returns input as output without change.

It does not introduce non-linearity.

Cell 4: Sigmoid Function

Code:

```
def sigmoid(x):
    return 1/(1+np.exp(-x))
```

Explanation (what to say in viva):

Sigmoid maps input to range 0 to 1.

It is useful for binary classification.

It introduces non-linearity.

Cell 5: Tanh Function

Code:

```
def tanh(x):  
    return np.tanh(x)
```

Explanation (what to say in viva):

Tanh outputs values between -1 and 1.

It is zero-centered.

It performs better than sigmoid in some cases.

Cell 6: ReLU Function

Code:

```
def relu(x):  
    return np.maximum(0, x)
```

Explanation (what to say in viva):

ReLU returns 0 for negative inputs.

It returns same value for positive inputs.

It is widely used in deep learning.

Cell 7: Softmax Function

Code:

```
def softmax(x):  
    e_x = np.exp(x)  
    return e_x/np.sum(e_x)
```

Explanation (what to say in viva):

Softmax converts values into probabilities.

All outputs sum to 1.

It is used for multi-class classification.

Cell 8: Plot Graphs

Code:

```
plt.plot(x, sigmoid(x))  
plt.plot(x, tanh(x))  
plt.plot(x, relu(x))  
plt.legend()  
plt.show()
```

Explanation (what to say in viva):

This plots multiple activation functions.

Different curves represent different functions.

Helps compare behavior visually.